

Automatic Detection and Segmentation of Tooth Cracks Based on Improved Mask R-CNN

Yuanchang Huang , Yadong Tang, Juhao Wu , Jiayan He , and Wenlong Wang 

Abstract—Early diagnosis and intervention of cracked teeth are crucial for preventing further dental damage. However, the detection of cracked teeth remains challenging for dental clinicians due to the subtle, complex, and irregular feature of these cracks. To address this issue, we propose an improved Mask R-CNN instance segmentation network for the automatic detection and segmentation of cracked teeth. Firstly, the backbone network was replaced with ResNeXt-50(32 × 4 d) to enhance the extraction of local features specific to cracks. Secondly, we introduced a Crack Feature Enhancement Module (CFEM), utilizing Bayesian optimization to fine-tune its hyperparameters, which leverages the pixel intensity differences between cracked and non-cracked regions to increase the sensitivity of the Feature Pyramid Network (FPN) to the complex features of cracks while suppressing irrelevant background information. Additionally, the mask head was redesigned into an encoder-decoder structure incorporating dynamic snake convolutions, which enables better capture of crack edge details and the integration of both deep and shallow feature information, with deep supervision applied to adjust the loss function weights. Extensive experiments and comprehensive evaluations demonstrate that our method outperforms the current state-of-the-art techniques. Furthermore, experiments on real intraoral images validate the effectiveness of our approach in detecting tooth cracks. Our model enables more accurate and earlier detection of cracked teeth, improving patient outcomes by allowing for timely

interventions, reducing the need for invasive treatments, and preserving dental structure.

Index Terms—Cracked teeth, improved Mask R-CNN, ResNeXt, crack feature enhance module, encoder-decoder mask head with dynamic snake convolutions.

I. INTRODUCTION

CRACK teeth are defined as teeth with fractures present in the vertical or horizontal direction involving the crown and/or extending to root structure [1], [2], [3]. Currently, problems associated with cracked teeth are one of the main causes of tooth loss [4], [5]. The presence of cracked teeth brings significant inconvenience and pain to individuals. Discomfort during chewing is the most common symptom of cracked teeth, and they are also particularly sensitive to temperature changes, especially cold [1], [6]. Furthermore, these cracks provide pathways for bacteria to invade the dental pulp. As bacteria colonize deeper into the cracks, the inflammatory response of the pulp becomes increasingly severe [5], [7]. Therefore, early diagnosis and intervention are essential to prevent further dental damage [2].

However, diagnosing cracked teeth in clinical trials is highly challenging due to the complexity of symptoms and the difficulty in locating cracks. Fig. 1 illustrates examples of cracked teeth. Fig. 1(a) shows a close-up of a tooth with a crack concealed in the grooves of the occlusal surface, making it easily overlooked. Similarly, Fig. 1(b) shows an overall image of a tooth with two cracks, one of which is very small and difficult to notice. These images vividly highlight the challenges in accurately detecting hidden tooth cracks. Traditional clinical examination methods rely on both the patient's symptoms and the dentist's observational experience. The former includes techniques such as occlusal testing [5], periodontal probing [5], staining [8], and cold stimulation [9], which can exacerbate symptoms and cause significant discomfort for the patient. The latter primarily involves transillumination [10], but the accuracy of this method is dependent on the dentist's diagnostic experience and current condition. With advancements in medical imaging technology, cone-beam computed tomography (CBCT) has gradually been adopted for the diagnosis of cracked teeth [11]. CBCT provides high-resolution three-dimensional images, aiding dentists in accurately locating and assessing cracks. Despite its high diagnostic accuracy, the high cost and radiation dose associated with CBCT remain significant considerations in its clinical application [11], [12].

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Scientific Ethics Committee of Guangzhou University and this study has been confirmed to fully comply with the Declaration of Helsinki and the relevant regulations of the People's Republic of China on biomedical human research. The approval number is GUSE[2024]095.

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Our code and datasets are available at <https://github.com/YCHuang18/ToothCrack>.

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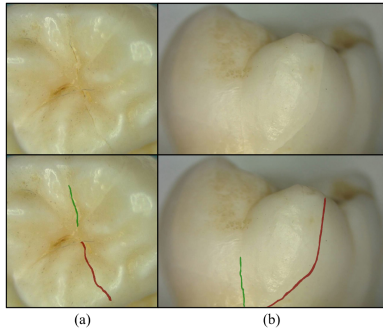


Fig. 1. Examples of cracked teeth. (a) localized image of a tooth's occlusal surface with cracks hidden in the grooves, (b) overall image of a tooth with tiny cracks.

In recent years, rapid advancements in computer vision technology in medical image analysis have offered a new possibility for detecting cracked teeth. Through deep learning, computer vision techniques can automatically analyze dental images to accurately detect and locate tooth cracks. Compared to conventional clinical methods and techniques aided by optical instruments, computer vision technology can not only enhance detection accuracy and efficiency but also reduce the reliance on clinical experience and minimize patient discomfort. Guo et al. [13] proposed an automatic tooth crack detection method based on optical images. This method combines the sliding window algorithm with convolutional neural networks to determine whether each window patch image contains a crack. Dumbryt et al. [14] introduced the use of X-ray tomography combined with CNNs for the non-destructive and qualitative analysis of tooth micro-cracks. These methods utilize deep learning models to automatically extract features from dental images, enabling precise diagnosis and localization. Among these computer vision techniques, instance segmentation offers a more refined solution compared to traditional object detection and semantic segmentation. It not only identifies each crack instance in the image but also precisely segments their contours and classifies them accordingly [15], [16], [17]. In the task of detecting and locating cracked teeth, this approach provides detailed information about the cracks, such as their exact location, direction, and severity.

Among instance segmentation models, Mask R-CNN stands out as one of the most classic. Its flexible and efficient instance-level recognition framework can be easily adapted to other tasks [18], making it a frequently cited benchmark model [19]. Mask R-CNN extends the Faster R-CNN [20] by incorporating an additional mask branch using a fully convolutional network (FCN) [21] to segment objects within detected bounding boxes [17]. Although Mask R-CNN has proven to be an effective instance segmentation method, there are still some limitations in its network structure and its application to crack detection tasks that warrant improvement. The issues with Mask R-CNN can be summarized as follows.

First, Mask R-CNN adopted the ResNet architecture as its backbone. ResNet achieves performance improvements by increasing network depth, but this comes at the cost of increased computational complexity, memory requirements, and training time as the network depth increases [22]. Secondly, in the

top-down propagation process of the Feature Pyramid Network (FPN) [23], only a 1×1 convolution was used to merge low-level and high-level feature maps. However, a 1×1 convolution adjusts only channel dimensions and fails to effectively utilize spatial information for feature enhancement, thereby lacking spatial processing capabilities. Moreover, cracks may appear in various orientations, and a single 1×1 convolution cannot effectively capture directional information to adapt to the orientation of cracks. Thirdly, Mask R-CNN's mask head applies only four consecutive 3×3 convolution operations to the feature maps from ROI Align. This approach may lead to inadequate fusion of global and local information, thereby impacting the segmentation accuracy [17], [24].

In response to the challenges faced by Mask R-CNN in detecting cracked teeth, this study proposes an enhanced model aimed at improving the performance of crack detection and achieving automated segmentation of cracked teeth in original intraoral images. Our improved model allows for more precise crack localization, reducing false positives and missed detections. This results in faster, more reliable diagnoses, better treatment planning, and ultimately improved patient care by minimizing the risk of tooth loss or complications from untreated cracks. Additionally, automated segmentation streamlines the diagnostic process, saving clinicians time and improving workflow efficiency. Specifically, our main contributions are outlined as follows:

- We replaced the original ResNet structure with ResNeXt to enhance the model's expressive power while maintaining lower computational complexity, thereby avoiding exponential growth in computation and memory.
- We have designed a crack feature enhancement module (CFEM) to assist Feature Pyramid Network (FPN) in enhancing crack features in different orientations during the fusion process. This module enables better capture of spatial information related to cracks.
- We proposed an encoder-decoder structure incorporating snake convolutions to capture both global information of the entire image and local details of cracks, thereby obtaining higher-quality crack masks.
- We created two experimental datasets for cracked teeth and conducted extensive experiments. Our findings demonstrate that our model not only produces high-quality masks but also achieves higher accuracy in instance classification.

II. RELATED WORK

A. Deep Learning-Based Dental Image Analysis

Deep learning has driven advancements in dental image analysis, making it possible to automatically extract complex patterns and features from dental medical images. Its applications in dentistry include detecting cavities, identifying periodontal disease, and segmenting dental structures from X-rays, CT scans, and intraoral images. Cui et al. [25] proposed a two-stage deep convolutional neural network named ToothNet for the automatic segmentation and identification of tooth instances from CBCT images, demonstrating outstanding performance on a self-constructed CBCT dataset. Casalegno et al. [26] developed

a semantic segmentation model to segment enamel, dentin, proximal caries, and occlusal caries from near-infrared transillumination images, achieving excellent segmentation results. Kühnisch et al. [27] employed MobileNetV2 for caries detection and classification, comparing their results with expert standards and achieving over 90% consistency. Zhou et al. [28] developed a separate scale-specific feature fusion network called WTNet for segmenting cone beam CT images of mandibular wisdom teeth in children. Experimental results showed that this network outperformed previous state-of-the-art segmentation methods. Deep learning has revolutionized dental image analysis, enabling accurate detection and segmentation of dental conditions from many optical imaging methods, including X-rays, CT scans, and intraoral images. It enhances diagnostic precision and efficiency, supporting tasks like cavity detection and dental structure segmentation. With advancements in imaging technology, deep learning continues to improve clinical workflows and diagnostic outcomes in dentistry.

B. Deep Learning-Based Cracked Tooth Detection

In recent years, deep learning techniques have been increasingly applied to dental imaging for the detection of cracked teeth, with the goal of assisting clinicians and reducing the rate of missed diagnoses. These approaches utilize a variety of imaging modalities, including panoramic X-rays, micro-CT, and optical microscope images. Fukuda et al. [29] evaluated the performance of the convolutional neural network DetectNet in detecting vertical tooth cracks in panoramic X-rays. Out of 330 vertical tooth cracks, the method successfully detected 267, but misdiagnosed 20 healthy teeth, leading to a relatively high misclassification rate for the classification task. Dumbryte et al. [30] applied the U-Net deep learning network to X-ray micro-CT images, successfully detecting both visible and hidden cracks in teeth. Their method, combined with 3D visualization, provided precise location information for micro-cracks. However, their method is complex and requires advanced expertise from dental professionals. Yang et al. [31] fine-tuned ResNet for the same task, achieving AUC values of 0.929 and 0.936, comparable to radiologists' performance. While effective, these methods focused on classification and lacked the ability to precisely localize cracks. Xie et al. [32] applied a semantic segmentation network based on DeepLabV3+, introducing dense linked dilated spatial pyramid pooling (Dense ASPP) and bottleneck attention modules (BAM) to predict tooth cracks in optical microscope images. Their method segmented crack regions effectively, but it struggled to distinguish multiple or overlapping cracks. These studies demonstrate promising directions for deep learning in cracked tooth detection but still face challenges such as limited localization accuracy, high complexity, and lack of generalization across clinical scenarios.

C. Instance Segmentation

In the past few years, instance segmentation technology has made significant progress and has become one of the important, complex, and challenging research directions in the field of computer vision. The goal of instance segmentation tasks is to

detect objects in images while precisely delineating the contours of each object. This technology finds broad applications in autonomous driving, robotic vision, medical image analysis, and other fields [33]. Among instance segmentation algorithms, Mask R-CNN [17] stands out as one of the most classic. Built on the Faster R-CNN object detection algorithm, Mask R-CNN adds a branch for segmentation masks, enabling the model to perform both object detection and high-precision instance segmentation. In addition to Mask R-CNN, many other instance segmentation models have been developed. Liu et al. [34] proposed PANet, which improves feature pyramids via path aggregation and adaptive pooling, enhancing segmentation in complex scenes. Wang et al. [35] introduced SOLO, reframing instance segmentation as per-pixel classification based on position and size, removing the need for region proposals while matching Mask R-CNN's accuracy. Tian et al. [36] developed the CondInst network, which employs conditional convolutions to generate specific convolution kernels for each instance, enabling flexible and precise segmentation. Chen et al. [37] proposed the BlendMask instance segmentation model, which adds a blender module to Mask R-CNN to better integrate global and local information, thereby improving segmentation accuracy. Additionally, the latest versions of the classic YOLO series object detection algorithms, such as YOLOv11 [38], have incorporated instance segmentation tasks, enhancing their versatility in handling both object detection and segmentation. By combining segmentation with object detection, these versions achieve real-time, efficient instance segmentation results. Despite these advances, most instance segmentation methods are designed with general-purpose datasets in mind and may not be directly applicable to domain-specific tasks. In particular, they often lack dedicated components for capturing subtle features, and can struggle with segmenting small-scale, low-contrast, or overlapping targets—common characteristics in many medical imaging scenarios.

III. METHODS

A. Overall Architecture

Fig. 2 shows the framework of our method. To better detect and segment tooth cracks, we have made specific improvements to address the structural deficiencies of Mask R-CNN. First, the original ResNet [39] backbone structure was replaced with ResNeXt [40], enhancing the model's representational power and performance while maintaining a lower computational complexity. The use of group convolutions in ResNeXt reduces the number of hyperparameters and avoids exponential growth in computation and memory usage. Second, we designed a crack feature enhancement module to fully exploit crack shape features and enhance the extraction of spatial information about cracks. This module compensates for the limitation of the FPN [23], which relies on a single 1×1 convolution to adjust feature channels during fusion. Lastly, an encoder-decoder mask head with dynamic snake convolutions was developed to replace the original Mask R-CNN mask head. This modification allows the generated masks to more accurately segment the cracked portions of teeth in the images, thereby improving the

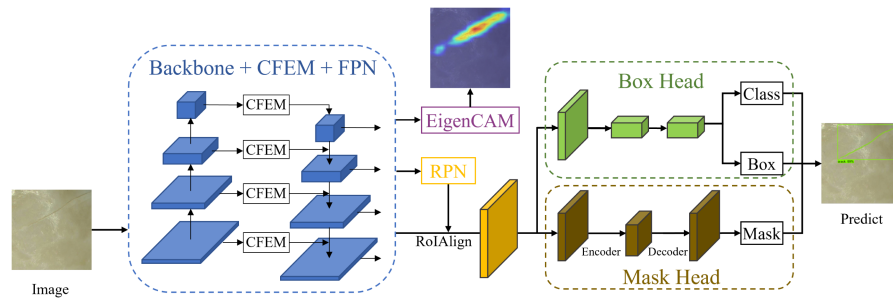


Fig. 2. Overall architecture of the proposed instance segmentation model for tooth crack detection. The input image is first processed by a feature extraction module composed of ResNeXt-50(32×4 d), CFEM, and FPN, generating multiscale feature maps (blue blocks). These features are visualized using EigenCAM to highlight backbone attention and simultaneously sent to the RPN to generate candidate regions (yellow blocks). The resulting features are then fed into the detection head (green blocks) for classification and bounding box regression, and into the segmentation head (brown blocks) to produce crack masks. Final outputs include class labels, bounding boxes, and segmentation results.

accuracy of pixel-level classification tasks. Additionally, we applied deep supervision in the mask head's loss function and used gradient-based methods to dynamically adjust the weights of each individual loss. Deep supervision enables the network to refine features across multiple layers, ensuring that intermediate outputs contribute to the final prediction. This approach enhances the model's ability to capture fine-grained details, leading to more precise segmentation of subtle and complex crack.

B. Backbone Based on ResNeXt

ResNeXt [40] is a modular architecture that enhances network representational capability by introducing the concept of cardinality, which is the size of the set of transformations. It follows a highly modular design akin to ResNet [39]/VGG [41] and uses grouped convolutions within the residual blocks to increase model capacity without raising computational costs. Each residual block in ResNeXt employs a combination of split, transform, and merge. (1) Splitting: the vector is sliced as a low-dimensional embedding, and in the above, it is a single-dimension subspace. (2) Transforming: the low-dimensional representation is transformed, and in the above, it is simply scaled. (3) Aggregating: the transformations in all embeddings are aggregated. Compared to the ResNet architecture, ResNeXt architecture excels in enhancing network expressiveness. Instead of merely increasing the network's depth or width, ResNeXt architecture augments its capacity by increasing the cardinality. This approach allows the model to capture more diverse features. Simultaneously, the use of grouped convolutions significantly reduces computational costs and improves computational efficiency. To ensure an appropriate network complexity and computational load, ResNeXt-50(32×4 d) was adopted as the backbone for our model, with the computational complexity of ResNet-50 serving as the baseline.

C. Crack Feature Enhance Module (CFEM)

The FPN [23] has demonstrated outstanding performance in multi-scale object detection tasks by effectively fusing feature maps of different scales, significantly improving detection accuracy. Through numerous experiments, the FPN structure has been proven effective in feature extraction. However, FPN

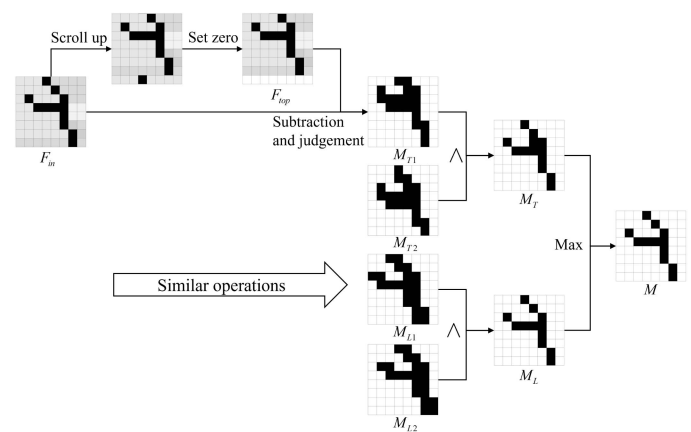


Fig. 3. Schematic diagram of the CFEM.

still has some limitations. Although the use of simple 1×1 convolutions for adjusting the number of feature map channels reduces computational complexity, it fails to fully exploit the spatial and directional information of cracks in images. This limitation can negatively impact its performance in the complex task of crack detection and segmentation. Cracks in dental images often exhibit distinct morphological features, characterized by abrupt changes in pixel intensity. Specifically, the pixels within a crack show significant differences in brightness and texture compared to the adjacent pixels. Given that cracks usually appear as elongated and irregular shapes, these spatial and directional characteristics are crucial for accurate crack detection. Moreover, some cracks can be challenging to detect due to the complex texture patterns in the background of the image, especially those on the tooth occlusal surfaces. Therefore, there is a need to design a crack feature enhancement FPN structure that fully leverages the spatial and directional characteristics of cracks, as well as the significant differences between the pixels within a crack and those on either side.

Based on the above analysis, we designed the Crack Feature Enhance Module (CFEM) as shown in Fig. 3. This module repeatedly utilizes the spatial and directional information of cracks by comparing the cracks with surrounding pixels or

feature points to enhance crack features and improve detection accuracy. CFEM processes the feature maps in both vertical and horizontal directions to generate crack masks, which are then combined with the input feature maps $F_{in} \in \mathbb{R}^{H \times W \times C}$. Specifically, to detect horizontal cracks, the feature map F_{in} obtained from the backbone network is first shifted by ω pixels along the vertical direction. To avoid confusion at the edges of feature map, the feature values that fall outside the boundary after shifting are set to 0. The specific formula is as follows:

$$F_{top}(i, j, c) = \begin{cases} F_{in}(i - \omega, j, c), & \text{if } \omega \leq i \leq H \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

$$F_{down}(i, j, c) = \begin{cases} F_{in}(i + \omega, j, c), & \text{if } 0 \leq i \leq H - \omega \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

where F_{top} and F_{down} represent the feature maps after scrolling upwards or downwards by a distance of ω pixels and then zero-padding, respectively; $F(i, j, c)$ represents the feature map F at the position with height i , width j , and channel c . Subsequently, by comparing the feature map before and after the shift, the longitudinal crack mask is generated. Specifically, the original input feature map F_{in} is compared with the feature maps (F_{top} and F_{down}) that have been shifted upward or downward and then zero-padded. The resulting difference represents the crack feature value, indicating the disparity between the crack and its surrounding areas. If this difference exceeds a predefined threshold θ , the location is identified as containing a crack, and the corresponding feature value is retained; otherwise, it is classified as background, and the feature value at that location is set to zero. The formula is as follows:

$$F_{T1} = \begin{cases} F_{in}(i, j), & \text{if } F_{in} - F_{up} > \theta \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

$$F_{T2} = \begin{cases} F_{in}(i, j), & \text{if } F_{in} - F_{down} > \theta \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where M_{T1} and M_{T2} represent the pixels feature maps that remain after subtracting the shifted feature maps from the original feature map and meeting the condition. A logical AND operation is then applied to combine M_{T1} and M_{T2} , removing feature points where the difference between the shifted values and the original feature map is non-zero but does not correspond to actual crack locations. The formula is as follows:

$$M_T = M_{T1} \wedge M_{T2} \quad (5)$$

where M_T represents the detected transverse crack feature map, and $\wedge$ denotes a logical AND operation that retains only the pixels simultaneously marked as 1 in both M_{T1} and M_{T2} . Similarly, to detect the longitudinal crack feature map M_L , the feature map is shifted horizontally. To account for the potential impact of diagonal and branching cracks on the detection results, we take the maximum value between the transverse and longitudinal crack feature maps. The formula is as follows:

$$M = \max(M_T, M_L) \quad (6)$$

Finally, to ensure consistent channel dimensions across feature maps, a 1×1 convolution is employed to adjust the number of

channels to 256. The output can be described as follows:

$$F_{out} = Conv_{1 \times 1}(M) \quad (7)$$

In the CFEM computation process, the width threshold ω and pixel threshold θ are two crucial hyperparameters that characterize the module's behavior and influence the network performance and detection results. For the width threshold ω , an appropriate setting can effectively capture the morphological features of cracks, making the distinction between cracked and non-cracked regions more pronounced. Regarding the pixel threshold θ , an optimal setting enables more accurate edge feature detection. Therefore, properly setting these two hyperparameters is essential for enhancing the network's detection performance. A series of experiments were conducted in subsequent trials to determine the optimal width threshold ω and pixel threshold θ , ensuring that the model's performance was maximized.

D. Mask Head

To achieve precise segmentation of crack images, the mask head needs to capture both global information of the image and local details of the cracks. In Mask R-CNN, the mask head processes the 14×14 feature maps obtained from RoI Align through four consecutive 3×3 convolutions, followed by a deconvolution operation to upsample the feature map to 28×28 for generating the final mask. He et al. [17] noted that the mask branch structure in Mask R-CNN is relatively simple and has significant room for improvement. Given that cracks are typically fine and morphologically complex structures, traditional convolution methods struggle to fully capture their detailed features. To better address this issue, a mask branch specifically tailored to more effectively capture and segment the intricate details of cracks was designed.

Based on the above analysis, our focus is on gathering useful global and detailed local information within the mask branch. To capture global information, an encoder-decoder network with skip connections was adopted, which has been proven effective in semantic segmentation tasks [42]. For capturing more detailed local information, dynamic snake convolutions were employed to further enhance the network. Consequently, we designed a mask head with an encoder-decoder structure that incorporates snake convolutions. The dynamic snake convolution, proposed by Qi et al. [43], is used for segmenting tubular structures. Its structure is illustrated in Fig. 4. The dynamic snake convolution process consists of two parts: (1) Bias computation: A convolution operation is performed on the input feature map to obtain offsets in the X and Y directions. These offsets are constrained within a specific range, allowing them to dynamically adjust the shape and position of the convolutional kernels by relying on the network's loss constraint during backpropagation. (2) Output feature map: The input feature map is convolved with the offset convolutional kernels to generate the output feature map. This specialized convolutional kernel is particularly effective for segmenting tubular structures. The dynamic snake convolution is then applied to the encoder-decoder mask head, as depicted in Fig. 5. The structure consists of an encoder on the left and a

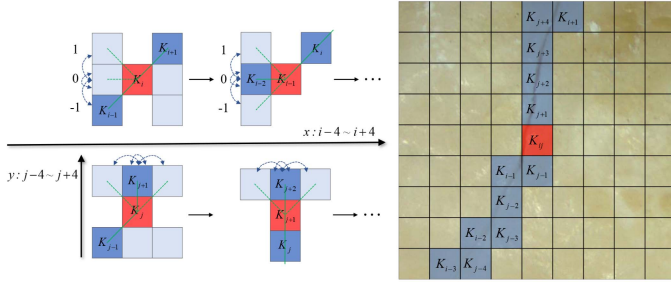


Fig. 4. Schematic diagram of the dynamic snake convolution. The red square represents the center pixel K_{ij} . Blue squares indicate the adaptively sampled neighbors. Blue dashed lines denote the learned offset paths. The green solid line shows the dominant crack direction, and the green dashed lines represent the extended receptive field along the crack.

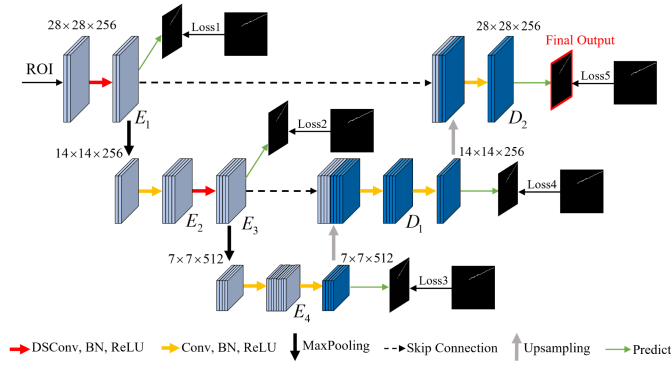


Fig. 5. Encoder-decoder structure incorporating dynamic snake convolutions. The dynamic snake convolution is represented in the figure as "DSCConv".

decoder on the right, with the encoder incorporating the dynamic snake convolution to more effectively extract crack features. Here, the size of the feature map I obtained through RoI Align is adjusted from 14×14 to 28×28 , and this serves as the input. First, the encoder performs snake convolution on the input feature map I . The operation process can be described as follows:

$$E_1 = DSCConv_{9 \times 9}(I) \quad (8)$$

where I represents the feature map input processed by RoI Align; $DSCConv_{9 \times 9}(\cdot)$ denotes a 9×9 dynamic snake convolution, followed by a batch normalization layer and a ReLU activation function (the latter two are not explicitly shown in the formula). Here, E_i indicates the i th processing of the feature map by the encoder. Subsequently, the feature map E_1 is further processed as follows:

$$E_2 = Conv_{3 \times 3}(Down(E_1)) \quad (9)$$

where $Down(\cdot)$ means maximum pooling of stride 2; $Conv_{3 \times 3}$ means a 3×3 convolution with stride 1, followed by a batch normalization layer and a ReLU activation function (the latter two are not explicitly shown in the formula). Here, the output channel of the encoder is doubled compared to the input feature

map E_1 . After that, the feature map E_2 processed by

$$E_3 = DSCConv_{9 \times 9}(E_2) \quad (10)$$

$$E_4 = Conv_{3 \times 3}(Down(E_3)) \quad (11)$$

where E_4 represents the final output of the encoder. Similarly, the output channels of the encoder here are quadrupled compared to the original input feature map I . Then our decoder first operates on the output E_4 of the encoder through

$$D_1 = Conv_{3 \times 3}(E_3 + Up(Conv_{3 \times 3}(E_4))) \quad (12)$$

where $Up(\cdot)$ means the bilinear interpolation operation and D_1 means the output of the first block of the decoder. Here, the number of channels is reduced to twice that of the original feature map E_4 . The same processing is then applied to D_1

$$D_2 = Conv_{3 \times 3}(E_1 + Up(Conv_{3 \times 3}(D_1))) \quad (13)$$

where D_2 is the final output of our mask head, with the size and number of channels of the output feature map D_2 being identical to those of the original input feature map I .

Compared to the original mask head in Mask R-CNN, which employs four consecutive convolutions, our structure has the advantage of capturing global information from the image while further extracting finer local details of the cracks, significantly enhancing the network's segmentation performance. Additionally, by using 28×28 resolution RoI feature maps instead of the original 14×14 resolution, we can better leverage local texture information.

E. Loss Function and Deep Supervision

The loss function is essential for optimizing the parameters within a model, as it quantifies the difference between the predicted results and the ground truth labels. By iteratively adjusting the model's weights, the loss function helps minimize the gap between the predicted values and the true values. During the training phase, this study employs a combined loss function incorporating bounding box regression, classification, and mask prediction branches to train the network. The loss of the network is defined as follows:

$$L = L_{cls} + L_{bbox} + L_{mask} \quad (14)$$

where L_{cls} is the loss of classification, L_{bbox} is the loss of bounding box regression, L_{mask} is the mask loss. The following are specific explanations of these types of losses. The definition of classification loss L_{cls} are as follows:

$$L_{cls} = - \sum_i \log [P_i P_i^* + (1 - P_i)(1 - P_i^*)] \quad (15)$$

where i is the index of an anchor in a mini-batch, P_i and P_i^* denote the predicted probability and ground truth of anchor, respectively. The ground-truth label P_i^* is 1 if the anchor is positive, and is 0 if the anchor is negative. The loss of bounding box regression L_{bbox} is defined as follows:

$$L_{bbox} = \frac{1}{N} \sum_{i=1}^N \sum_{j=\{x,y,w,h\}} \text{smooth}_{L_1}(t_{ij} - t_{ij}^*) \quad (16)$$

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2, & \text{if } |x| > 1 \\ |x| - 0.5, & \text{otherwise} \end{cases} \quad (17)$$

where $t = (t_x, t_y, t_w, t_h)$ is a vector representing the 4 parameterized coordinates of the predicted bounding box, and $t^* = (t_x^*, t_y^*, t_w^*, t_h^*)$ is that of the ground-truth box associated with a positive anchor. To achieve precise segmentation of crack images, the mask loss was improved using a deep supervision approach. Auxiliary losses were added at multiple points within the mask branch, allowing the model to iteratively refine feature representations at different levels, thereby enhancing segmentation accuracy. Specifically, as shown in Fig. 5, the feature maps output at these points are used to compute auxiliary mask losses $L_{mask,m}$ through image prediction. These losses are measured using binary cross-entropy to quantify the difference between the predicted mask and the ground truth mask. Assuming the mask output at the m th layer is M_m and the ground truth mask is M_m^* , the mask loss at the m th layer can be expressed as:

$$L_{mask,m} = -\frac{1}{k^2} \sum_{1 \leq i,j \leq k} [M_{m,ij}^* \log(M_{m,ij}) + (1 - M_{m,ij}^*) \log(1 - M_{m,ij})] \quad (18)$$

where $M_{m,ij}$ and $M_{m,ij}^*$ represent the predicted and ground truth values of the j th pixel in the i th sample, respectively, and k denotes the height or width of the mask. Notably, when $m=5$, the mask output at this layer is the final prediction. The overall mask loss can then be defined as follows:

$$L_{mask} = \sum_{m=1}^5 \lambda_m L_{mask,m} \quad (19)$$

where λ_m represents the weight corresponding to the m th loss. To balance the mask loss with other losses and simplify the hyperparameter tuning process, the sum of the weights in the mask loss was set to 1. For the individual losses within the mask loss, we adopt a gradient adjustment method. The principle of this method has already been proven effective in tuning other hyperparameters. For example, Zhi et al. [44] introduced an efficient fine-tuning technique that leverages gradient-based parameter selection. Their research demonstrated that, by adjusting only a small subset of parameters from a pre-trained model while freezing the rest, it's possible to achieve performance on par with or even superior to full model fine-tuning. In our approach, gradient adjustment works as follows: after each training batch, the loss function generates gradients, which are then used to guide the parameter updates. These gradients are then used to adjust and normalize each loss branch, as described in the following equations:

$$\lambda_m \leftarrow \lambda_m - \eta \frac{\partial L_{mask,m}}{\partial \lambda_m} \quad (20)$$

$$\lambda_m \leftarrow \frac{\lambda_m}{\lambda_{\text{sum}}} \quad (21)$$

where η is the learning rate, λ_{sum} means the sum of the weights for each loss in the mask loss. Specifically, the mask branch generates a $k \times k$ mask for each RoI and each category. With categories, the total output is $n \times k \times k$. For the predicted binary

mask output, the Sigmoid function is applied to each pixel, and the result is used as the input for the binary cross-entropy loss function. The overall loss is defined as the average binary cross-entropy loss.

IV. DATASETS AND IMPLEMENTATION

A. Creation and Labeling of Datasets

An ideal dataset should contain a large number of high-resolution intraoral images of cracked teeth, covering various types, and severities of cracks, with detailed and high-quality annotations. However, obtaining such a dataset presents numerous challenges. For example, collecting a sufficient number of clinical images of cracked teeth requires access to extensive case data; capturing high-quality intraoral images necessitates professional equipment and techniques; and the task of labeling the dataset is extremely labor-intensive. Given these difficulties, extracted human teeth were opted to be used, and cracks were created artificially through thermal expansion and contraction, followed by detection on real cracked teeth in patients' mouths. In the process of creating the dataset, we collected 432 freshly extracted human molars and artificially generated cracks using the thermal expansion method described in the literature [45]. These 432 cracked teeth served as the basis for acquiring high-resolution images under clinical imaging setups, providing the foundation for dataset construction. This controllable method enables the generation of realistic features, and expert inspection confirmed their clinical resemblance, helping to reduce potential bias. This study was reviewed by the Scientific Ethics Committee of Guangzhou University and has been confirmed to fully comply with the Declaration of Helsinki and the relevant regulations of the People's Republic of China on biomedical human research. The approval number is GUSE[2024]095.

Subsequently, images of the cracked teeth were captured using an optical microscope equipped with an industrial camera, at a resolution of 1920×1080 pixels. From these, a total of 832 images with cracks of suitable size were selected. To ensure the model focuses on the details of the cracks during training, the original high-resolution images (1920×1080 pixels) were cropped into 1,370 images of 512×512 pixels, which served as the primary training dataset. This dataset was named Dataset A, with "crack" as the sole detection target. To maintain sufficient detail while providing a large enough field of view to capture the overall appearance and distribution of cracks on the teeth, the original images were also cropped into 840 images of 1200×800 pixels, naming this dataset Dataset B. The detection targets in Dataset B are categorized based on the positions of cracks relative to the teeth, including "horizontal crack", "vertical crack", "oblique crack", "branching crack", and "occlusal crack". Both datasets were divided into training, validation, and test sets in an 8: 1: 1 ratio. The details of the two datasets are shown in Table I. For reproducibility purposes, the two annotated tooth crack datasets used in this study are publicly accessible at <https://github.com/YCHuang18/ToothCrack>. To expand the dataset and enhance the model's robustness, data augmentation techniques were applied to the training sets, including horizontal flipping, vertical flipping, and random brightness adjustments.

TABLE I
DETAILED INFORMATION OF DATASETS

Dataset	Number of images	Resolution	Instance type	Number of instances
Dataset A	1370	512×512	crack	1754
Dataset B	840	1200×800	horizontal crack	230
			vertical crack	262
			oblique crack	193
			branching crack	189
			occlusal crack	212

Both datasets were annotated by a dental expert to ensure the accuracy and consistency of the labeling process. The expert's involvement was particularly important for accurately labeling cracks that were subtle or hidden, ensuring high-quality ground-truth for training the model.

B. Implementation Details

The experiments were conducted on a hardware platform consisting of an Intel(R) Xeon(R) Gold 6130 CPU, 25 GB of memory, and an NVIDIA Tesla V100 GPU with 32 GB of memory. The model was trained and tested on a Linux (Ubuntu 20.04) system, using PyTorch 2.0.0, Python 3.8, and CUDA 11.8.

We initialized the model weights using a pre-trained model on the COCO dataset [46] to accelerate the training process. Subsequently, we trained the improved network on labeled cracked tooth images. Drawing from our training experience and considering the GPU power limitations, we determined the following hyperparameters to achieve optimal and stable training performance: a learning rate ranging from 0.01 to 0.0001 with a cosine annealing decay strategy, a batch size of 8, stochastic gradient descent (SGD) optimizer with a momentum of 0.9, weight decay of 0.0001, and a total of 100 epochs. The selection of the learning rate and batch size was thoroughly tested in supplementary experiments, which guided our final choices.

V. EXPERIMENTS AND EVALUATIONS

A. Evaluation Metrics

The standard COCO evaluation metrics [46] were used on Dataset A and Dataset B, including AP^{bbox} , AP_{50}^{bbox} , AP_{75}^{bbox} , AP^{seg} , AP_{50}^{seg} and AP_{75}^{seg} . AP summarizes the shape of the precision-recall curve and is defined as the average precision at a set of evenly spaced recall levels. The AP^{bbox} / AP^{seg} represents the average of these 10 tests for bounding box detection/segmentation, which is used as the final value for each 0.05 change from IoU=0.5 to IoU=0.95. The AP_{50}^{bbox} / AP_{50}^{seg} obtained by setting the IoU threshold to 0.5 for bounding box detection/segmentation, and the AP_{75}^{bbox} / AP_{75}^{seg} represents the mAP obtained by setting the IoU threshold to 0.75.

B. Hyperparameter Selection

To achieve optimal performance for the network model, a series of experiments were conducted to compare various hyperparameters on Dataset A. These included the choice of width

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT WIDTH AND PIXEL THRESHOLDS IN CFEM USING GRID SEARCH AND BAYESIAN OPTIMIZATION

Type	Width	Pixel	AP^{bbox}	AP_{50}^{bbox}	AP_{75}^{bbox}	AP^{seg}	AP_{50}^{seg}	AP_{75}^{seg}
Grid search	3	0.20	76.99	94.48	84.23	44.72	88.84	38.00
	5	0.30	77.33	94.64	84.02	44.83	88.30	38.88
	7	0.40	77.21	95.11	84.45	44.86	89.59	37.55
	9	0.50	76.69	95.15	83.70	44.68	89.56	36.74
	11	0.60	78.10	95.84	85.69	45.48	90.05	39.01
Bayesian optimization	8	0.46	79.51	95.75	86.78	47.80	91.79	42.71

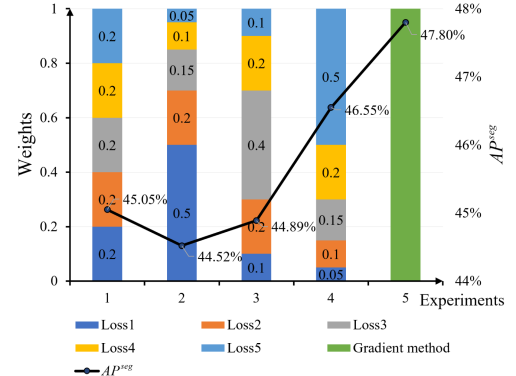


Fig. 6. Comparison of deep supervision models with different weights. Experiments 1 to 4 are fixed weight combinations, and Experiment 5 is the result of using the gradient adjustment method with Experiment 4 as the initial weights.

and pixel thresholds in the CFEM and the weight selection for different loss functions in the deep supervision. The details of these experiments are presented below.

Hyperparameter Selection Experiments in CFEM: Detailed experiments and optimization were conducted on two key hyperparameters of the CFEM: the width threshold ω and the pixel threshold θ . To find the optimal parameter combination, both grid search and Bayesian optimization methods [47] were used. As shown in Table II, a grid search was conducted with parameter combinations: (3, 0.2), (5, 0.3), (7, 0.4), (9, 0.5), and (11, 0.6), based on an analysis of crack morphology. Width thresholds between 3 and 11 were considered effective for capturing crack features without adding too much background noise, while pixel thresholds between 0.2 and 0.6 aimed to balance noise and crack detection. To refine the parameters, bayesian optimization was applied, running 100 iterations within the specified ranges. The optimal combination found was (8, 0.46). Among these threshold combinations, the set (8, 0.46) identified through Bayesian optimization exhibited the best performance. Therefore, a combination of a width threshold of 8 and a pixel threshold of 0.46 was selected as the parameters for the CFEM in subsequent experiments.

Deep Supervision Model With Different Weights: Experiments were conducted to compare different weight combinations for the loss functions in deep supervision. As shown in Fig. 6, we tested several combinations with varying emphasis on the loss functions in the mask head, including uniform distribution, higher weights for early-stage losses, higher weights for

TABLE III
PERFORMANCE COMPARISON WITH DIFFERENT BACKBONES

Backbone	AP^{bbox}	AP_{50}^{bbox}	AP_{75}^{bbox}	AP^{seg}	AP_{50}^{seg}	AP_{75}^{seg}
ConvNeXt-tiny	63.30	93.95	72.07	29.54	83.67	11.59
Swin-T-tiny	32.72	67.08	27.38	16.37	57.59	3.30
PVT-v2-b1	55.77	89.47	57.83	26.98	79.37	9.01
ResNet-50	77.66	94.53	83.81	44.72	89.39	37.04
ResNeXt-50(32×4d)	79.51	95.75	86.78	47.80	91.79	42.71

late-stage losses, and configurations with higher weights for specific stages. The results showed that combinations with higher weights for late-stage losses performed the best. This is because Loss5, being calculated based on actual output and the real dataset, should have a larger weight to achieve better network performance. Additionally, to achieve a higher AP^{seg} , the loss should gradually decrease as the network deepens. Taking the weight combination of the fourth experiment in Fig. 6 as the initial weights, we further optimized this optimal combination using the gradient adjustment method. As shown in the fifth experiment in Fig. 6, Compared to the model with initial weights, the adjusted gradients resulted in a notable increase in the AP^{seg} of model by 1.25%, demonstrating a significant improvement. This experiment demonstrated the importance of appropriately adjusting the weights of loss functions in deep supervision to enhance model performance, especially highlighting the effectiveness of gradient adjustment in optimizing weight distribution and thus improving detection accuracy.

C. Ablation Studies

In this study, a series of experiments were conducted on Dataset A, including a comparison of different backbone networks and an analysis of the effects of various improvement steps. These experiments aimed to demonstrate the significant performance enhancements resulting from the replacement of the backbone network and the modifications applied at each step.

Backbone Comparison: To demonstrate the effectiveness of replacing the backbone network with the ResNeXt architecture, five different backbones were compared: ResNet-50 [39], ResNeXt-50(32 × 4 d) [40], ConvNeXt-tiny [48], Swin-transformer-tiny [49], and PVT-v2-b1 [50]. These backbones represent CNN and transformer architectures in computer vision. They are characterized by similar complexity and floating-point operations, ensuring a fair comparison in this experiment. Table III presents the quantitative results for these backbones. The results demonstrate that the model employing ResNeXt-50(32 × 4 d) as the backbone yields slightly higher performance compared to that using ResNet-50, whereas the remaining three backbones exhibit markedly inferior results in both bounding box detection and segmentation tasks, with Swin-Transformer-Tiny performing the worst. The superior performance of ResNeXt-50 and ResNet-50 can be attributed to their convolutional architectures, which are well-suited for capturing local spatial features, essential for detecting subtle tooth cracks. In contrast, the other backbones, relying on global

self-attention mechanisms and non-overlapping patch tokenization, may struggle to preserve fine-grained local details, limiting their effectiveness in identifying narrow, low-contrast cracks. Based on the experimental results and architectural advantages, ResNeXt-50(32 × 4 d) emerges as the most suitable backbone for the proposed framework. To gain deeper insights into the performance of different backbones in tooth crack detection, we employ Eigen-CAM [51] to generate heatmaps that clearly depict the attention distribution of each network when processing sample images, with these results included in the supplementary materials.

Effect Analysis about Various Improvements: To validate the effectiveness of the various network improvements, the ablation studies were conducted based on the baseline Mask R-CNN to better understand the impact of each modification. Specifically, comparative experiments were performed on the following five aspects: (1) replacing the backbone network with ResNeXt-50(32 × 4 d), (2) incorporating the CFEM in the FPN, (3) using an encoder-decoder structure in the mask head, (4) employing snake convolutions in the mask head, and (5) applying deep supervision in the loss function. As shown in Table IV and V, improvements in model performance are demonstrated by the results of index 1, 2, 4, 5, 6, & 10 in Table IV, where each modification contributed to the overall enhancement. From the comparison of index 1 & 2, index 7 & 9, and index 8 & 10 in Table IV and index 1 & 2 in Table V, it is evident that replacing the backbone network with ResNeXt not only enhances feature extraction capability but also reduces model complexity, as indicated by the decrease in the number of parameters (44.28 M → 43.44 M) and FLOPs (341.41 G → 331.43 G). This demonstrates that ResNeXt effectively balances performance and computational efficiency. The addition of the CFEM module (index 3 in Table V) does not result in a substantial increase in overall complexity, while providing a modest improvement in segmentation accuracy (42.37 → 42.53). This highlights CFEM's high cost-benefit ratio in enhancing crack feature representation without imposing a substantial computational burden. On this basis, the introduction of the encoder-decoder structure and dynamic snake convolution (index 4 in Table V) leads to an increase in parameters and FLOPs. However, the performance gains, particularly in segmentation accuracy (AP: 42.37 → 47.80), justify this increase. As indicated by the comparison between index 4 and index 10 in Table V, the encoder-decoder structure enhances mask detail fidelity, while dynamic snake convolution optimizes crack edge detection, significantly boosting segmentation performance. Despite the increase in complexity, detection speed remains within an acceptable range, suggesting a reasonable trade-off between accuracy and computational cost. Future work will focus on reducing the complexity associated with dynamic snake convolution to further improve model efficiency and deployment performance in practical applications. To provide a more intuitive visualization of the contributions of each component across multiple evaluation metrics, radar charts have been plotted and included in the Supplementary Material. Overall, these enhancements significantly improve the model's ability to detect and segment complex crack patterns

TABLE IV
ABLATION STUDY RESULTS ON THE IMPACT OF NETWORK IMPROVEMENTS FOR CRACK DETECTION AND SEGMENTATION

Index	ResNeXt	CFEM	Encoder-decoder	DSConv	Deep supervision	AP^{bbox}	AP_{50}^{bbox}	AP_{75}^{bbox}	AP^{seg}	AP_{50}^{seg}	AP_{75}^{seg}
1						77.66	94.89	83.73	42.37	87.25	34.85
2	✓					77.64	95.00	85.00	42.88	89.14	35.42
3		✓				77.24	94.88	84.75	42.53	88.43	36.25
4	✓	✓				77.65	95.72	85.73	43.33	86.74	37.25
5	✓	✓	✓			76.87	94.62	84.05	45.92	90.21	40.56
6	✓	✓	✓	✓		79.31	95.62	85.62	47.72	90.35	43.57
7			✓	✓	✓	77.97	95.71	85.49	46.00	90.60	41.36
8		✓	✓	✓	✓	78.21	95.78	85.53	46.61	89.23	41.89
9	✓		✓	✓	✓	78.96	95.59	85.57	47.42	90.34	43.37
10	✓	✓	✓	✓	✓	79.51	95.95	86.78	47.80	91.79	42.71

TABLE V
IMPACT OF MODEL MODIFICATIONS ON COMPLEXITY

Index	ResNeXt	CFEM	Encoder-Decoder with DSConv	Params	FLOPs	FPS
1				44.28M	341.63G	24.61
2	✓			43.44M	331.43G	24.96
3	✓	✓		43.44M	331.43G	24.96
4	✓	✓	✓	62.27M	479.22G	18.88

and subtle cracks, reinforcing the effectiveness of the proposed modifications in challenging dental environments.

D. Comparisons With Other Methods

In this study, comparative experiments were conducted on two datasets, Dataset A and Dataset B, to demonstrate that our network model outperforms several popular instance segmentation models in the detection and segmentation of cracked teeth. All models use the same learning rate, batch size, optimizer and epoch settings and take the best training weights for testing.

Comparative Study on Dataset A: To validate the effectiveness of our model, we compared it against several popular instance segmentation models on Dataset A, including Mask R-CNN, BlendMask, CondInst, and YOLOv11. These models were selected as benchmarks due to their outstanding performance in instance segmentation tasks and their widespread application in various complex visual tasks. Specifically: (1) YOLOv11 [38] is highly regarded for its lightweight and efficient design, particularly excelling in object detection tasks; (2) BlendMask [37] effectively combines high-level and low-level semantic information to extract more accurate instance segmentation features, making it well-suited for segmentation tasks in complex and dynamic scenes.; (3) CondInst [36] introduces dynamic weight generation, aiming to enhance the flexibility of instance segmentation; (4) Mask R-CNN [17] is a classic instance segmentation framework known for its stability and strong performance, making it a standard benchmark.

Table VI presents the quantitative results of these models on the Dataset A test set. We can observe that, under relatively lenient conditions, our model lags behind BlendMask in bounding box detection (the AP_{50}^{bbox} of our model is lower than that of BlendMask) and behind CondInst in segmentation tasks (the

TABLE VI
PERFORMANCE COMPARISON WITH OTHER METHODS ON DATASET A

Method	AP^{bbox}	AP_{50}^{bbox}	AP_{75}^{bbox}	AP^{seg}	AP_{50}^{seg}	AP_{75}^{seg}	FPS
YOLOv11 [38]	74.10	94.63	82.54	39.26	92.84	29.58	30.2
BlendMask [37]	78.07	97.01	84.45	30.03	83.40	6.48	16.2
CondInst [36]	78.35	96.87	78.26	43.42	95.86	25.90	22.8
Mask R-CNN [17]	77.66	94.89	83.73	42.37	87.25	34.85	18.4
Ours	79.51	95.75	86.78	47.80	91.79	42.71	16.5

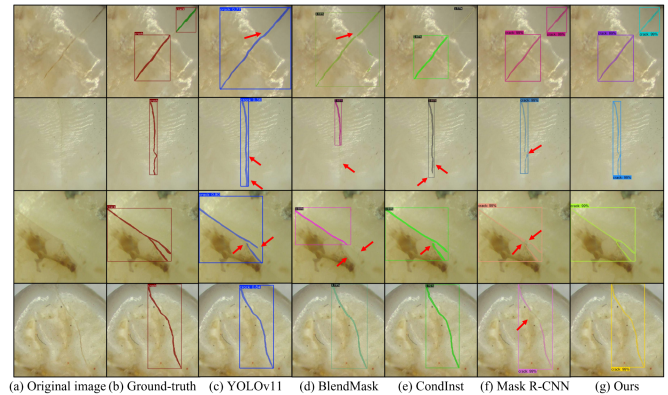


Fig. 7. Comparison of tooth crack instance segmentation visualization results for different models on dataset A. (a) original image, (b) ground-truth, (c) YOLOv11, (d) BlendMask, (e) CondInst, (f) Mask R-CNN, (g) our model.

AP_{50}^{seg} of our model is lower than that of CondInst). However, under more stringent conditions, our network demonstrates superior precision, as evidenced by significantly higher AP_{75}^{bbox} and AP_{75}^{seg} compared to the other models. Consequently, the overall precision of our model surpasses that of these benchmark models. In general, our model exhibits higher accuracy in detecting crack boundaries, enabling more precise localization and identification of cracks. Although our model operates at 16.5 FPS, which is slightly slower than Mask R-CNN yet faster than BlendMask, the performance remains within a practical range for near-real-time deployment, particularly in clinical scenarios where high segmentation accuracy is of greater importance.

The visual comparison results are shown in Fig. 7. Our model demonstrates outstanding performance on Dataset A. Compared to other methods, it achieves clear and complete segmentation of

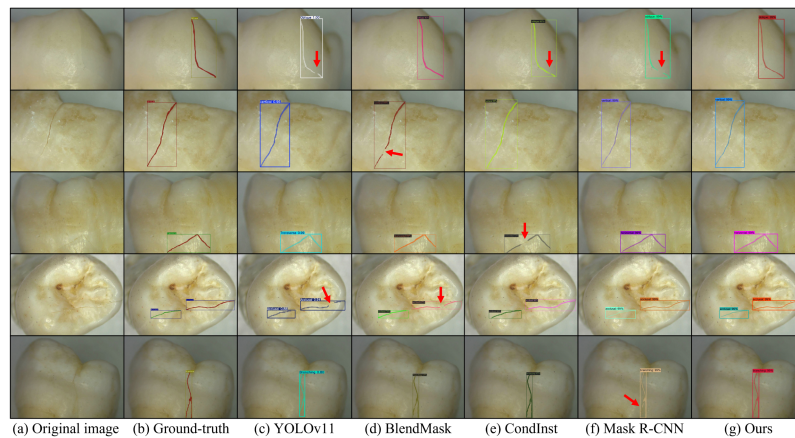


Fig. 8. Comparison of tooth crack instance segmentation visualization results for different models on Dataset B. (a) original image, (b) ground-truth, (c) YOLOv11, (d) BlendMask, (e) CondInst, (f) Mask R-CNN, (g) our model.

TABLE VII

PERFORMANCE COMPARISON WITH OTHER METHODS ON DATASET B

Method	AP^{bbox}	AP_{50}^{bbox}	AP_{75}^{bbox}	AP^{seg}	AP_{50}^{seg}	AP_{75}^{seg}	FPS
YOLOv11 [38]	74.84	95.67	74.85	37.35	91.82	18.96	25.3
BlendMask [37]	74.96	97.85	92.20	26.29	75.29	6.33	13.7
CondInst [36]	79.02	99.00	94.81	43.11	94.52	20.72	20.1
Mask R-CNN [17]	85.04	99.05	98.39	43.18	94.31	38.31	16.3
Ours	90.72	99.49	98.70	47.02	96.04	41.88	14.8

tooth crack contours, with results closely aligning with manual annotations. This notable advantage is likely attributed to several enhancements made to the mask head, including the implementation of an encoder-decoder structure and snake convolutions. These improvements significantly enhance the detail fidelity of segmentation masks and edge detection capabilities. Through these optimizations, our model is better able to capture the contours and details of complex and fine cracks, resulting in superior segmentation performance.

Comparative Study on Dataset B: The experiments were conducted on Dataset B, which focuses on detecting cracks at various locations on teeth. As with the models compared on Dataset A, we evaluated Mask R-CNN, BlendMask, CondInst, YOLOv11, and our own network model on Dataset B. Table VII presents the quantitative results of these models on the Dataset B test set. The results indicate that our network model significantly outperforms these benchmark models in both bounding box detection and segmentation performance. It is noteworthy that, compared to the results on Dataset A, the segmentation accuracy on Dataset B has decreased slightly, while the overall precision of the bounding boxes has markedly improved. This phenomenon may be attributed to the relatively smaller size of the cracks in the images of Dataset B, which makes each pixel's correct classification have a more pronounced effect on overall segmentation accuracy, leading to a decrease in segmentation precision. On the other hand, the accuracy of bounding box detection mainly depends on the discernibility of the crack's start and end points. Cracks in Dataset B are relatively easier to distinguish at these points compared to those in Dataset A,

resulting in improved bounding box precision. On Dataset B, our model runs at 14.8 FPS, which is slightly lower than on Dataset A due to the larger input image size, but the speed remains within a usable range for practical deployment. Additionally, we conducted an F1-score experiment for popular models on single images from the test set of Dataset A and Dataset B, which are presented in the supplementary materials, further demonstrating the superior performance of our model.

The visual comparison results are shown in Fig. 8. Our model exhibits exceptional performance on Dataset B. In the visualizations of other methods, it is evident that their predictions often exhibit some degree of pixel discontinuity in the cracks. In contrast, our segmentation results closely align with the manual annotations.

E. Predicted Results in Various Environments

Predictive Results on Challenging Images: To comprehensively evaluate the performance of our model in various complex environments, we tested it on images that are challenging to observe with the naked eye. As shown in Fig. 9 these challenging conditions include poor lighting, low light, very narrow and concealed cracks, and cracks located in the grooves of the tooth's occlusal surface. The reason for selecting these environments for testing is that they often represent scenarios in clinical practice where dentists find it difficult to detect cracks, thus placing higher demands on the model.

In terms of instance segmentation performance, our model's predictions are closely aligned with manual annotations under these complex conditions. As shown in Fig. 9, in environments with poor or low lighting, the model continues to accurately locate and segment the crack regions, demonstrating its robustness in low-light conditions. For cracks located in the grooves of the tooth's occlusal surface, the model effectively captures crack information within complex backgrounds, reflecting its capability to recognize fine and intricate structures. Additionally, when detecting narrow and concealed cracks, the model demonstrates high precision and can identify micro-cracks that are difficult for the human eye to discern. These experimental

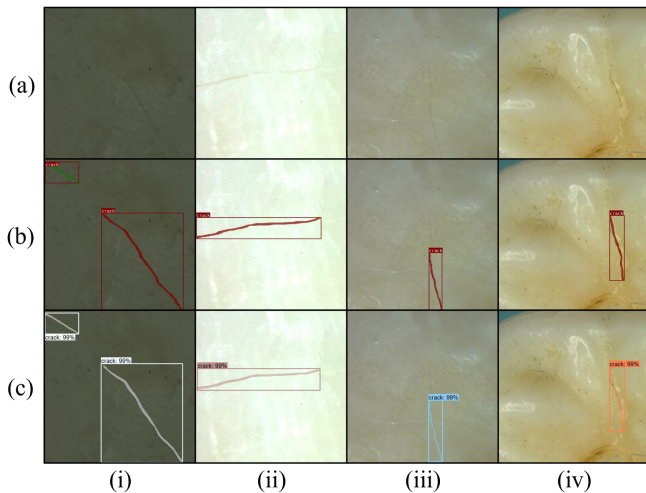


Fig. 9. Visualization of tooth crack prediction based on our model in various environments. (a) original image, (b) ground-truth, (c) our model, (i) low light environment, (ii) high light environment, (iii) very narrow and concealed cracks (iv) cracks located in the grooves of the tooth's occlusal surface.

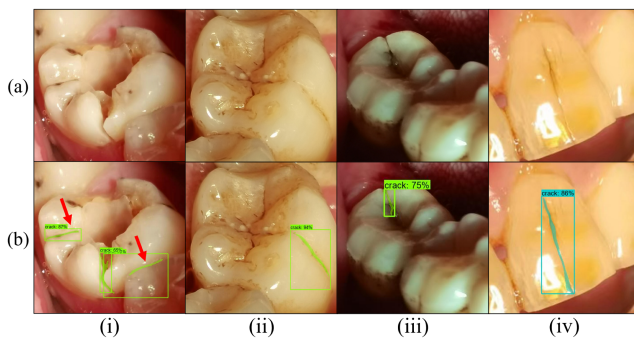


Fig. 10. Prediction results of our model in intraoral cracked tooth images. (a) original images, (b) predicted results. (i)–(iv) represent real intraoral images containing cracked teeth.

results indicate that our model not only performs exceptionally well under normal conditions but also provides reliable detection and segmentation results in a variety of challenging environments, further validating its potential and practicality for real applications. Furthermore, we have conducted comparative experiments on our method before and after improvement under different disruptive environments, which are presented in the supplementary materials, further proving the robustness of our approach.

Prediction results in intraoral cracked tooth images: To further evaluate the model's performance on real intraoral tooth images, we used the weights trained on Dataset A to make predictions. Fig. 10 presents the prediction results of our model on clinical intraoral images of tooth cracks. These results were validated by professional dental doctors to ensure accuracy. Overall, the model accurately identified tooth cracks in most images. However, in Fig. 10(i), there were some instances of misidentification where the model mistakenly identified gaps between teeth and a pit-and-fissure caries as cracks. The tooth

cracks in the remaining images were correctly recognized. These misidentifications may be due to the visual similarities between tooth gaps and cracks, as well as the similar appearance of pit-and-fissure caries and cracks in the images. Overall, the prediction results are acceptable.

VI. DISCUSSION

The motivation behind this study is to develop customized deep learning methods tailored to address the unique challenges presented by crack image features in tooth crack datasets, particularly for crack detection and instance segmentation tasks. Based on these tailored approaches, a series of experiments were conducted to validate their effectiveness. The experimental results can be discussed from the following perspectives.

Backbone network built from ResNeXt structures: As representatives of computer vision tasks, CNN and transformer architectures are highly popular, primarily due to their exceptional feature extraction capabilities. CNNs excel at capturing local features and patterns, offering good translational invariance, making them suitable for a variety of image recognition tasks. On the other hand, transformers utilize self-attention mechanisms to capture feature relationships on a global scale, providing strong contextual understanding, and have demonstrated outstanding performance in various image-related tasks in recent years. In this study, the selected representative CNN and transformer backbone networks exhibited different performance levels in the crack detection task. In the Class Activation Mapping (CAM) experiments with backbone networks, the ResNet and ResNeXt architectures showed more focused attention on crack locations compared to the other three backbones. The reason behind this is that while ConvNeXt architecture introduces modern convolutional network design elements, such as larger kernels and layer normalization, these improvements may be better suited for capturing larger-scale global features and complex high-level semantics. However, crack detection relies primarily on small, local features. Similarly, the transformer-based backbone networks, such as PVT-v2-b1 and Swin-transformer-tiny, also excel at capturing global features. Through self-attention mechanisms, they can establish feature relationships over a broader range. Nevertheless, these types of networks lack the ability to effectively capture local features such as cracks, and they often require significantly more computational resources and longer training times. The ResNet and ResNeXt architectures are specifically designed with an emphasis on local feature extraction, which is crucial for identifying fine, localized features such as cracks. ResNeXt, in particular, enhances ResNet's capabilities by incorporating grouped convolutions, thereby increasing the diversity and richness of the extracted features. This design improvement makes from this analysis, we can conclude that architectures like ResNeXt, which prioritize local feature extraction, are better suited for the demands of crack detection and segmentation.

Effectiveness of target-specific feature enhancement modules: Tooth cracks typically exhibit fine, complex, and irregular patterns, which pose significant challenges for detection tasks. By employing a pixel-contrast approach, the CFEM calculates the pixel-level differences between crack and non-crack regions,

generating an enhanced feature map that emphasizes the edges and details of cracks. This enhancement makes the model more sensitive and accurate in crack detection by leveraging the subtle pixel-level differences between cracks and the surrounding healthy tissue, allowing the model to better capture the intricate morphology of cracks. Therefore, to enhance the detection of certain describable target features, module design can be tailored to the identifiable image characteristics of the detection target. For instance, specialized feature extraction and enhancement can be developed for attributes such as shape, edges, and texture. This target-feature-based module design not only improves detection accuracy but also enhances the model's adaptability to complex scenarios.

Effectiveness of the Encoder-Decoder Mask Head with Dynamic Snake Convolutions: The dynamic snake convolution technique allows for adaptive adjustment of the convolutional kernel's shape and position, enabling more precise capture of crack edges and details. This flexible convolution approach is particularly well-suited for handling the complex and irregular morphology of cracks, leading to superior performance in segmentation tasks. Traditional convolution operations often face limitations when dealing with edges and fine details, but dynamic snake convolutions overcome this by adjusting the kernel based on input image features, thereby enhancing edge detection and improving the precision of feature extraction. The encoder-decoder structure further bolsters the model's ability to extract and integrate features, allowing it to retain more spatial information, refine segmentation boundaries, and enhance the quality of segmentation details. The results demonstrate that the design of flexible convolutional kernels has significant potential in handling complex edges and fine features, while the encoder-decoder structure's advantage in high-resolution feature recovery underscores the importance of multi-scale feature fusion in improving segmentation accuracy.

Although the experimental results show promising performance of our customized methods, the current model still exhibits limitations in overall complexity and operational efficiency. Future work will focus on the following areas: 1) A more diverse and comprehensive dataset will be developed to capture features from different perspectives and enhance crack detection across various cases. Future work will focus on expanding the dataset to include varying crack morphologies, severity levels, and dental surface regions, improving model generalization and robustness in clinical scenarios. 2) Efficient network optimization techniques will be applied to reduce model complexity while retaining strong feature representation. To ensure practical deployment, efforts will be directed at minimizing model parameters and computational overhead through methods such as pruning, quantization, and knowledge distillation, thereby enhancing detection efficiency without sacrificing accuracy. 3) The model will be integrated into medical imaging devices to facilitate real-time, multi-angle dental crack detection. This integration will enable seamless deployment on intraoral scanners and dental imaging equipment, providing real-time support for clinical diagnosis. Additionally, mobile deployment on edge devices like smartphones and tablets will improve accessibility, streamlining workflows and enhancing patient outcomes.

VII. CONCLUSION

In the task of tooth crack detection and segmentation, we propose an improved Mask R-CNN instance segmentation deep learning framework. Compared to other models such as CondInst, BlendMask, and traditional Mask R-CNN, our approach integrates the ResNeXt backbone network, which enhances the model's focus on local features of cracks. Additionally, the CFEM emphasizes the subtle differences between crack and non-crack regions, improving the model's sensitivity to the complex characteristics of cracks. To further enhance the CFEM's effectiveness, its design is optimized by Bayesian optimization to identify the best combination of hyperparameters. We further enhance edge segmentation capabilities by incorporating dynamic snake convolutions in the mask head with an encoder-decoder structure. In parallel, gradient adjustment methods are used to fine-tune the weights of the mask branch loss function, further refining the model's accuracy in detecting cracks. Experimental results demonstrate that, compared to the classic Mask R-CNN, this improved approach shows superior performance in both detection and segmentation tasks, as evidenced by higher AP metrics. Overall, the proposed Mask R-CNN improvement method shows significant potential in enhancing the accuracy and reliability of tooth crack detection and segmentation. We believe that our work will prove invaluable to the research community for detecting and segmenting tooth cracks.

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